

VPA IB DP CS — Topic 7: Control (HL Only, Weeks 1 – 9)

Topic 7: Control 控制 Kòngzhì HL ONLY

IB Syllabus: Topic 7 (7.1 – 7.3) | Level: HL ONLY | Duration: 9 weeks (30 hours)

Topic 7.1: Centralized Control Systems 集中式控制系统 Jízhōngshì Kòngzhì Xìtǒng

Week 1 — Introduction to Control Systems 控制系统简介

Lesson	Content	Activities	Differentiation
1	Review of control systems from Topic 6: sensors, actuators, feedback (复习传感器、执行器、反馈)	Concept map linking Topics 6 and 7	All: Contribute; Gold: Identify abstractions across both topics
2	Centralized control: single controller, all decisions made centrally; examples (traffic control centre, factory SCADA)	Case study: city traffic management centre	Bronze: Identify components; Silver: Explain flow; Gold: Analyse single point of failure
3	Communication in centralized systems: polling, master-slave, request-response protocols	Diagram communication patterns; compare efficiencies	All: Diagram; Gold: Calculate communication overhead
4	Advantages and disadvantages of centralized control; bottleneck, latency, reliability concerns	Debate: centralized vs emerging alternatives	All: Participate; Gold: Reference CAP theorem conceptually
5	TOK: Centralized control concentrates power (and knowledge) in one place. Is this an epistemic advantage (one clear picture) or a vulnerability (one point of failure)?	Discussion; written argument	All: Contribute; Gold: Link to political philosophy

TOK Knowledge Question: A centralized controller has a "God's eye view" of the system it controls. Is complete information always possible, or do centralised systems necessarily operate with simplified models? What are the epistemic limits of centralized control?

Week 2 — Centralized System Design and Reliability

Lesson	Content	Activities
1	System modelling for centralized control: state machines, decision tables	Design state machine for elevator control

Lesson	Content	Activities
2	Redundancy in centralized systems: hot standby, cold standby; failover mechanisms	Design failover for hospital life support
3	Latency and real-time constraints: hard vs soft deadlines; jitter	Calculate worst-case response times
4	Case study: air traffic control — centralized coordination of aircraft	Analyse ATC as centralized system; identify constraints
5	Formative: Centralized control system analysis (20 marks)	Case study response

Topic 7.2: Distributed Systems 分布式系统 Fēnbùshì Xìtǒng

Week 3 — Distributed Control Fundamentals 分布式控制基础

Lesson	Content	Activities
1	Distributed systems: multiple autonomous nodes cooperating; no single point of control (分布式系统, 节点, 自治)	Compare to centralized using Venn diagram
2	Communication in distributed systems: message passing, shared memory, RPC, publish-subscribe	Diagram each pattern; match to use case
3	Consensus problems: agreement, validity, termination; Byzantine generals problem (conceptual)	Simulation: reaching consensus with faulty nodes
4	Distributed algorithms: leader election, mutual exclusion (Ricart-Agrawala conceptual)	Trace leader election on small network
5	TOK: In a distributed system, no single node has complete knowledge. Can the system as a whole be said to "know" things that no individual node knows?	Discussion; connection to emergent properties

TOK Knowledge Question: Distributed systems exhibit emergent behaviour that no single node is aware of. Is this emergence a form of collective knowledge, or merely the appearance of knowledge? Can a system "know" something without any part of it knowing it?

Week 4 — Distributed System Challenges

Lesson	Content	Activities
1	CAP theorem: consistency, availability, partition tolerance; trade-offs	Analyse systems: which two does each choose?
2	Fault tolerance: replication, checksums, heartbeat, gossip protocols	Design fault-tolerant key-value store conceptually
3	Clock synchronisation: logical clocks, Lamport timestamps (conceptual), physical clock challenges	Why does order matter in distributed transactions?

Lesson	Content	Activities
4	Case study: blockchain as distributed consensus mechanism	Analyse blockchain through CAP lens
5	Formative: Distributed system design scenario (25 marks)	Design + justify distributed solution

Week 5 — Cloud and Edge Computing 云计算与边缘计算 Yún Jìsuàn yǔ Biānyuán Jìsuàn

Lesson	Content	Activities
1	Cloud computing as distributed system: service models, deployment models revisited	Analyse VPA's cloud dependencies
2	Edge computing: processing near data source; latency reduction; bandwidth saving	Compare cloud vs edge for IoT, autonomous vehicles
3	Microservices architecture: decomposition, inter-service communication, containers	Diagram microservice architecture for e-commerce
4	Load balancing: round-robin, least connections, geographic; CDN	Design CDN for global video streaming
5	Consolidation: Centralized vs distributed comparison essay	8-mark timed response

Topic 7.3: Autonomous Agents 自主代理 Zìzhǔ Dàilǐ

Week 6 — Autonomous Systems 自主系统 Zìzhǔ Xìtǒng

Lesson	Content	Activities
1	Autonomous agents: perception, decision-making, action; sense-plan-act cycle (感知-规划-执行)	Map sense-plan-act to robot vacuum cleaner
2	Reactive vs deliberative architectures: Braitenberg vehicles, subsumption architecture	Simulate reactive behaviours; compare to planned
3	Multi-agent systems: cooperation, competition, emergent behaviour, swarm intelligence	Watch swarm robotics videos; model simple rules
4	Machine learning in autonomy: supervised, unsupervised, reinforcement learning (conceptual)	Discuss: can an agent learn without being programmed?
5	TOK: An autonomous agent makes decisions without human intervention. Who is responsible when it makes a mistake? Is the agent a moral agent, or merely a tool?	Structured debate; prepare arguments

TOK Knowledge Question: If an autonomous vehicle must choose between hitting a pedestrian or swerving into a wall (harming its passenger), how should it decide? Is this a question of programming, ethics, or both? Can there be a "correct" answer, or is any choice merely a programmed preference?

Week 7 — Ethical and Social Implications

Lesson	Content	Activities
1	Autonomous weapons: lethal autonomous weapons systems (LAWS); international law	Research UN positions; prepare position statement
2	Surveillance and privacy: facial recognition, predictive policing, mass data collection	Case study: smart city surveillance in Shenzhen
3	Algorithmic bias: training data bias, feedback loops, fairness definitions	Analyse bias in given datasets; discuss mitigation
4	Employment and economic impact: automation, job displacement, universal basic income debate	Policy proposal: manage transition to automation
5	Formative: Ethical evaluation of autonomous system (20 marks)	Written response with balanced argument

Week 8 — Case Study Integration (Paper 3 Preparation)

Lesson	Content	Activities
1	Case study methodology: how to approach Paper 3	Review past case studies; identify question types
2	Case study deep reading: annotating, question generation, concept linking	Annotate case study; generate 5 exam questions
3	Connecting case study to syllabus: where do Topics 1–7 apply?	Concept map: case study → syllabus links
4	Research skills for case study: evaluating sources, citing evidence	Source evaluation exercise; bibliography practice
5	Mock Paper 3 question practice	Timed response; peer assessment

Week 9 — Topic 7 Assessment and Consolidation

Lesson	Content	Activities
1	Revision: Centralized, distributed, autonomous — comparison framework	Framework construction; rote vs understanding
2	Revision: Ethical implications across all three system types	Ethical scenario cards
3	Assessment: Topic 7 End-of-Topic Test (1 hour, 50 marks)	Formal conditions
4	Test feedback; DIRT	Markscheme analysis
5	Case study progress check; IA final submission reminder	Individual tutorials

Key Vocabulary (Topic 7)

English	中文	Pinyin	Definition
Centralized control	集中式控制	jízhōngshì kòngzhì	A system where all decisions are made by a single controller
Distributed system	分布式系统	fēnbùshì xìtǒng	A system with multiple autonomous nodes cooperating
Consensus	共识	gòngshí	Agreement among distributed nodes on a single value
Autonomous agent	自主代理	zìzhǔ dàilǐ	An entity that perceives its environment and acts independently
Emergent behaviour	涌现行为	yǒngxiàn xíngwéi	Complex behaviour arising from simple interactions
CAP theorem	CAP定理	CAP dìnglǐ	A distributed system can guarantee at most two of: consistency, availability, partition tolerance
Failover	故障转移	gùzhàng zhuǎnyí	Automatic switching to a redundant system upon failure
Swarm intelligence	群体智能	qúntǐ zhìnéng	Collective behaviour of decentralized, self-organized systems
Edge computing	边缘计算	biānyuán jìsuàn	Processing data near the source rather than in a central cloud
Algorithmic bias	算法偏见	suànfǎ piānjiàn	Systematic errors in algorithmic outcomes due to biased data or design

Teacher Checklist — End of Topic 7:

- ☐ All IB 7.1 – 7.3 syllabus points covered
- ☐ Students can compare centralized, distributed, and autonomous systems
- ☐ Students can evaluate ethical implications of autonomy
- ☐ Case study annotations completed by all HL students
- ☐ IA final submission uploaded to IBIS
- ☐ TOK discussions documented
- ☐ Victoria Park Academy Shenzhen | IB DP Computer Science | Topic 7 Scheme of Work